



Region-specific LOI–TOC conversion models for marine sediment blue carbon assessment in the Western Irish Sea

Anthony Grey · Ciara Ní Mhaoldomhnaigh ·
Mark Chatting · Phoebe Walsh · Alan Lee ·
Brian Kelleher · Mark Coughlan

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Abstract Continental-shelf sediments (<200 m water depth; ~9% of the global seafloor) store large amounts of organic carbon (OC) and accumulate OC annually, but management and accounting applications require regional evidence and transparent uncertainty. Loss-on-ignition (LOI) is a rapid, low-cost proxy for organic matter and OC, but LOI-to-total organic carbon (TOC) conversions vary with sediment type and analytical protocol. This study used multiagency Western Irish Sea data to test whether region- and sediment-specific LOI-to-TOC regressions improve OC estimation relative to a pooled conversion. Paired LOI and TOC measurements were strongly correlated ($R^2=0.948$). Structural-break and analysis of covariance (ANCOVA) tests identified changes in the

LOI–TOC relationship associated with mud content, iron (Fe) concentration and water depth. Splitting samples into high-mud and low-mud groups produced group-specific models ($R^2=0.97$ and 0.72 , respectively) and improved fit for high-mud sediments, while the low-mud model retained greater uncertainty. The results support the use of separate sediment-class-specific LOI-to-TOC calibration models for Western Irish Sea sediments and comparable settings, provided that paired LOI–TOC calibration, the range of sediments to which each model applies, and propagated uncertainty are reported. By integrating archived samples, routine geotechnical measurements, grain-size data, LOI results and targeted TOC analyses from multiple survey and research sources, the study also demonstrates how existing marine monitoring and seabed-mapping

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A. Grey (✉) · C. N. Mhaoldomhnaigh · B. Kelleher
School of Chemical Science, Dublin City University,
Dublin, Ireland
e-mail: greyanthony@gmail.com

C. N. Mhaoldomhnaigh
e-mail: ciara.nimhaoldomhnaigh22@mail.dcu.ie

B. Kelleher
e-mail: brian.kelleher@dcu.ie

M. Chatting
School of Civil Engineering, University College Dublin,
Dublin, Ireland
e-mail: M.Chatting@exeter.ac.uk

P. Walsh · M. Coughlan
School of Earth Sciences, University College Dublin,
Dublin, Ireland
e-mail: phoebe.walsh@ucdconnect.ie

M. Coughlan
e-mail: mark.coughlan@ucd.ie

A. Lee
Dún Laoghaire-Rathdown County Council, Marine Road,
Dún Laoghaire, Dublin, Ireland
e-mail: alan.lee93@gmail.com

M. Coughlan
Research Ireland Centre for Applied Geosciences
(iCRAG), O'Brien Centre for Science East, University
College Dublin, Dublin, Ireland

activities can be leveraged to strengthen national sediment carbon datasets.

Keywords Blue carbon · Marine sediment carbon assessment · Organic carbon · Loss-on-ignition (LOI) · Sediment classification · Continental shelf sediments

Introduction

Subtidal, non-vegetated marine sediments hold a major fraction of ocean OC, with the top metre estimated to contain > 2.3 Tt OC and accumulate 126–350 Mt OC yr⁻¹ (Atwood et al., 2020; Epstein & Roberts, 2022). Continental shelves cover ~9% of the global marine area and include OC-rich sediments that can hold ~2% total organic carbon (TOC) and are exposed to climate and anthropogenic pressures (Diesing, 2020; Epstein et al., 2025; Legge et al., 2020; Smeaton & Austin, 2022; Smeaton et al., 2020). Fine-grained shelf deposits, including those in the Irish Sea, Fladden Grounds and Celtic Sea, are recognised OC accumulation hotspots (Roseby et al., 2026; Ward et al., 2025). In these areas, OC-rich sediments are found in low-energy depositional settings where fine-grained sediments composed primarily of silts and clays (i.e. < 63 µm), are allowed to settle from suspension and accumulate on the seafloor. Their carbon content reflects mineralogical and geochemical controls, including clay minerals and Fe oxides that can protect organic matter (Babakhani et al., 2025; Kennedy & Wagner, 2011; Moore et al., 2023; Zhang et al., 2025).

Reliable OC measurements are needed to support stock estimates, spatial predictive models and monitoring. To address this, collaboration among researchers, policymakers, and international organizations is essential to develop and implement consistent approaches and standardized methodologies for assessing and reporting OC stock, accumulation and burial rates, thereby supporting the development of a rigorous evidence base. Mapping studies depend on local measurements propagated into unsampled areas using statistical or machine-learning approaches (Chatting et al., 2025; Diesing, 2020; Diesing et al., 2017; Wilson et al., 2018). Although databases such as the Modern Ocean Sediment Archive and Inventory of Carbon (MOSAIC),

EURO-CARBON and PANGAEA improve data access (Felden et al., 2023; Graversen et al., 2025; Paradis et al., 2023), inconsistent methods for OC stock, accumulation and burial-rate estimation still limit comparability across studies (Dahl et al., 2025; Graves et al., 2022; Ratefinjanahary et al., 2025).

Direct OC measurement by combustion analysis after inorganic carbon removal is widely used and can be accurate and reproducible when preparation, acidification and calibration are controlled (King et al., 1998; Serrano et al., 2023; Wang et al., 2011). However, elemental analysis is relatively costly and requires specialist equipment, consumables and routine calibration, which can restrict high-density monitoring. This is especially relevant for agencies with limited budget and developing nations. Alternative approaches include Walkley–Black wet oxidation, which can overestimate OC in anoxic sediments and underestimate it in oxic sediments (Leong & Tanner, 1999; Tuffour et al., 2014). LOI is widely used by researchers, monitoring agencies and commercial laboratories as a semi-quantitative OC proxy (Breithaupt et al., 2019; Hoogsteen et al., 2015; Plater et al., 2015) and is included in BS EN 15935:2021 for soil, waste, treated biowaste and sludge (British Standards Institution, 2021). LOI measures mass loss during ignition and can include organic matter (OM), structural water, salts and carbonate-related losses; therefore, conversion to OC requires local calibration.

Published LOI-to-TOC conversions differ among arid soils, mangroves and littoral sediments (Jensen et al., 2018; Ratefinjanahary et al., 2025; Touch et al., 2017), partly because sediment composition, OM content and ignition temperature affect regression slopes (Sleutel et al., 2007; Breithaupt et al., 2023; Table S3). Ireland's Exclusive Economic Zone (EEZ) covers roughly 450,000 square kilometres (~10 times its land mass), yet its seabed carbon inventory remains poorly constrained despite the extensive degree of mapping and sampling that has occurred there (Chatting et al., 2025). Here, we compile LOI, TOC, grain-size and iron (Fe) data for Western Irish Sea (WIS) non-vegetated marine sediments from research, commercial and national monitoring sources, and test whether stratified LOI-to-TOC models provide a practical framework for regional OC assessment. We also provide recommendations for methodological aspects of sediment carbon analysis and recommendations for

national carbon stock accounting strategies which can inform interested stakeholders.

Regional setting

The Western Irish Sea study area (Fig. 1) connects to the Atlantic Ocean through the North Channel and St. George's Channel (Knight et al., 1999). Sample sites span mud, sand and gravel sediments and were retrospectively grouped in Fig. 1 as High mud ($> 19.80\%$ mud) or Low mud ($\leq 19.80\%$ mud) as described by the results of statistical testing from this study.

Water depths range from coastal shallows (< 5 m) to approximately 115 m in the central basin. The western region is generally deeper, less tidally energetic and more saline (Gowen & Stewart, 2005), supporting a seasonal summer gyre and offshore Summer Stratified Region (Woods et al., 2019; Fig. 1) distinct from the Coastal and Southern Mixed Regions (Hill et al., 1994, 1997; Horsburgh et al., 2000; O'Reilly et al., 2014). North of 53.5°N , weaker hydrodynamics favour fine-grained deposition (i.e. silts and clays), whereas southern areas are dominated by sands, gravelly sands and high-energy bedforms (Coughlan et al., 2021; Creane et al., 2022; Trimmer et al., 2003; Ward et al., 2025). The Irish Sea catchment is $\sim 43,000$ km², with major freshwater input in the eastern Irish Sea (Bowden, 1980). Previous work documented anthropogenic impacts on Western Irish Sea Mud Belt sediments, including trawling and radionuclide inputs from Sellafield (Coughlan et al., 2015; Muir et al., 2026).

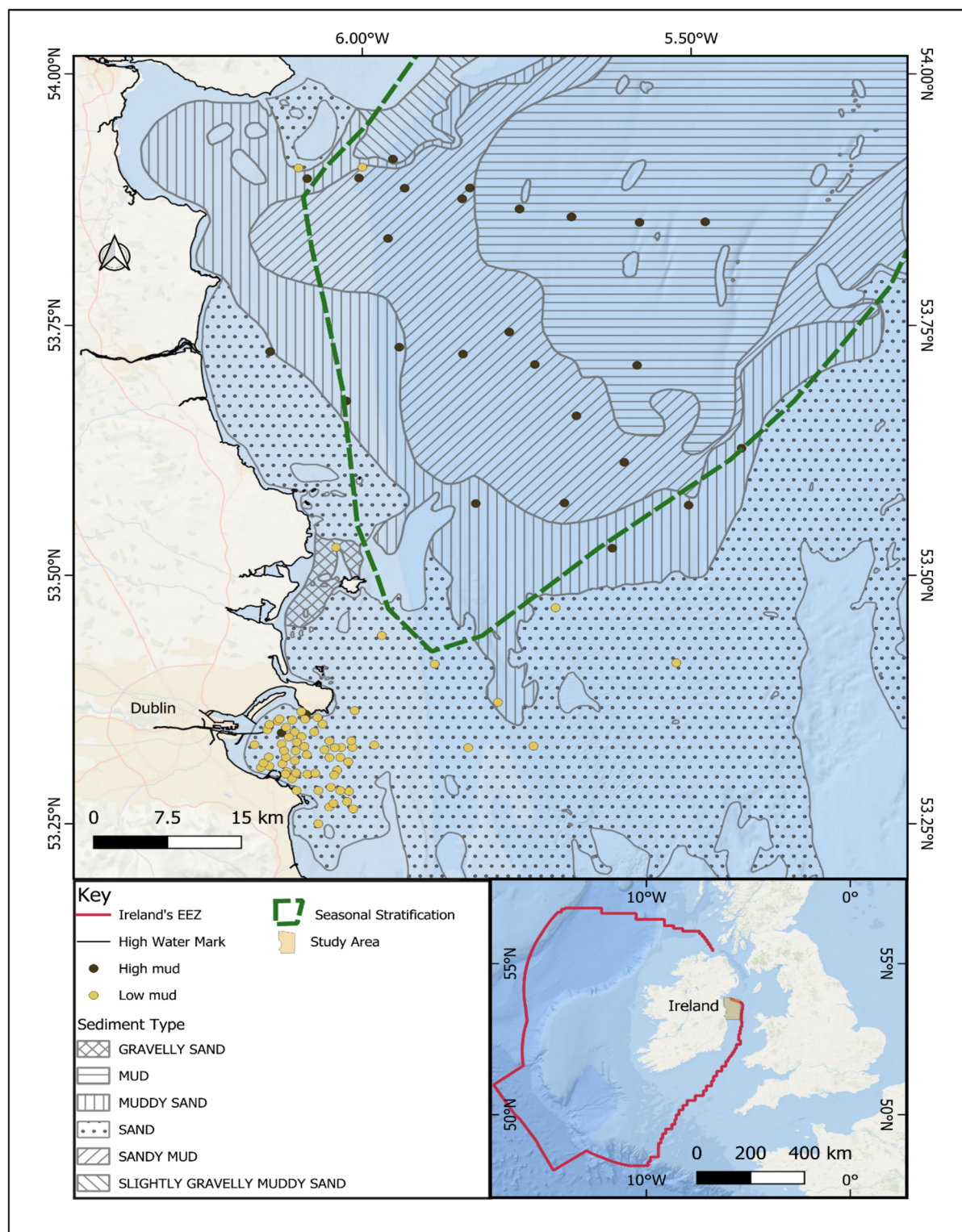
Data and methods

Samples and existing data were collated from surveys between 2010 and 2023: CV10_28 (O'Reilly et al., 2010), KRY22_03 (Furey & Cullen, 2022) and CE23_016 (QUEST; grant no. PBA/CC/21/01). Data variables and source agencies are summarised in Table 1. Where data had been published previously, the relevant reference is provided. CV10_28 surface sediments were sampled in June 2010 during INFOMAR (Integrated Mapping for the Sustainable Development of Ireland's Marine Resource)

survey CV10_28 aboard the RV Celtic Voyager using a Day Grab. Box cores were taken using a Reineck box corer, and subsequently push cores were taken from the box using pre-cut and edge filed piping (50 cm length and 7 cm internal diameter), capped, sealed and frozen at -20°C . For survey KRY22_03, the Marine Institute (MI) collected surface grab samples from Dublin Bay in spring 2022 using a Day Grab aboard the RV Keary (Geological Survey Ireland), and samples were refrigerated until analysis. In October 2023, the QUEST project team led survey CE23_016 aboard the RV Celtic Explorer (MI) to execute a sampling programme in the Western Irish Sea. Surface grab samples were captured using a Day Grab and stored at -20°C . A Reineck box corer was used similarly to survey CV10_28 where duplicate push cores (hereafter referred to as box cores) were extracted from the sediment box on deck before storage (1* frozen). Gravity (1–3 m) and Vibrocores (4–5.5 m) were used to collect deeper sediments where sediments were collected within plastic piping inserted into respective coring devices before deployment to the seabed. Onboard cores were sectioned into 1 m lengths before storage at -20°C . The processing of frozen cores was carried out at Dublin City University. Workflow of grab samples included defrosting, complete homogenisation using a stainless-steel trowel before subsampling for respective analysis. The box cores, gravity cores, and vibrocores were split manually while frozen using an angle grinder power tool. Briefly, the two deep lines were scored along the surface of the piping at opposing sides. A flat-blade screwdriver and hammer were used to gently tap along the score lines until the outward pressure split the pipe to reveal the frozen and physically intact core. The outer core was QC assessed for any possible contaminants from the pipe. Stainless steel saws were used immediately while the core remained frozen to further section the core into selected horizons before subsampling.

Sediment analysis

Grain size analysis was carried out on the 0–5 cm horizon of core samples and on the homogenised subsample from grabs (sand, silt, and clay fractions). Grain size was measured on the < 2 mm fraction of



◀**Fig. 1** Map of the Western Irish Sea study area and sample locations. Seabed sediment classification is derived from the British Geological Survey Seabed Sediments 250 K dataset (British Geological Survey, 2011; Folk, 1954). Sample sites are shown as High mud (> 19.80% mud) and Low mud (≤ 19.80% mud). Approximate extent of the summer seasonal water mass stratified region is adapted from Woods et al. (2019)

dried and sieved sediment by laser diffraction (Malvern Mastersizer MS3000) after dispersion in deionised water and sonication. The fraction retained in the sieve (> 2 mm) was weighed and classified as gravel mass, later integrated into the data set during processing with software. All studies used the same methodology to attain grain size distribution data. Data was subsequently processed using the statistics package GRADISTAT in Excel to define sediment grain size distribution and provide % of sand, silt, clay, and gravel (Blott et al., 2001). Mud (%) was derived from the sum of silt and clay. CV10_28 samples were analysed at University College Cork; KRY22_03 samples were analysed by Fugro (<https://www.fugro.com/expertise/laboratory-testing-and-data-delivery>), and CE23_016 samples were analysed at University College Dublin.

TOC was measured consistently across surveys on the < 2 mm sediment fraction after freeze-drying, powdering by manual grinding with a mortar and pestle, and storage in a desiccator. A 125 µm sieve was used only after powdering samples as a post grain size test to ensure that all powdered sediments passed the same size cutoff to obtain a uniform fine fraction prior to weighing, i.e. sediment that did not pass through was further powdered and combined with the rest of the sample before homogenising. CV10_28 samples were analysed at the University College Dublin Microanalytical Laboratory KRY22_03 and CE23_016 samples were analysed in the Dublin City University environmental laboratory. TOC concentrations are reported as weight percent of dry sediment (wt.%). TOC data used in this study for CV10_28 was carried out on the 0–2 cm depth horizon of cores, while cores analysed for this study were sampled 0–3 cm to increase sample availability. CE23_016 core sediments used 0–5 cm depths. Briefly, powdered sediment samples were weighed into silver capsules (~ 10 mg) using a calibrated balance (QUINTIX35-1S- readability 0.01 mg). Sediments were treated to remove carbonate (Verardo et al., 1990). The serial addition (20 µl*5)

of 1 M HCl was pipetted into the silver capsule until, followed by drying overnight at 40 °C, and storage in a desiccator until analysis. Immediately prior to analysis, silver capsules were closed and lightly compacted using stainless steel tweezers and analysed by combustion using a Fisons NCS 1500 NA elemental analyser, calibrated with a certified soil standard (CE Instruments; mean C = 2.3%, % RSD < 1%) before each batch and after every nine samples. % RSD for sample triplicates ranged from 1.02 to 5.5 with the higher variability occurring in sediments with higher sand and shell content.

To perform LOI analysis, sediment samples (~ 40–50 g wet weight) were placed on glass petri dishes and dried at 105 °C for 24 h to remove all moisture, until constant mass was achieved. Dried sediments were sieved to < 2 mm and manually powdered using a mortar and pestle. Samples were re-dried at 60 °C for an additional 4 h, cooled, and stored in a vacuum desiccator prior to ignition (ensuring complete dryness after processing). Replicate sample masses of 10–15 g were placed in pre-weighed ceramic crucibles. The samples were furnace at 450 °C for 4 h, allowed to cool in the furnace, and immediately placed in a desiccator until re-weighing for mass loss. LOI (wt.%) was calculated as: $LOI (wt.%) = (Y - X/Y) \times 100$ where, Y is the initial dry sediment mass, and X is the post-ignition mass (determined as total crucible + ash mass—crucible tare). Replicate analyses and desiccation were used to reduce precision and rehydration artefacts. In most cases duplicate analysis (some samples had triplicate runs) was carried out for LOI with % RSD ranging from 1.3% in homogenous mud sediments to 13.6% in heterogeneous sandy and lower—OM sediment.

Fe concentrations were measured in triplicate using a Thermo Scientific Niton XL3t 900 portable X-ray fluorescence (XRF) analyser with Niton NDTTr 6.5.2 software (Radu et al., 2013). Quantification used fundamental parameters to reduce matrix effects, with internal calibration before and after each run. Accuracy was checked against certified soil reference materials (NCSDC 73308, TILL-4 and GBW 07411) following US EPA Method 6200 (U.S. EPA., 2007).

Statistical analysis

Data were quality checked in Excel and analysed using Python statsmodels (Seabold & Perktold,

Table 1 Summary of agencies and sources providing data and samples for this publication

Survey	Agency	Date (m/yr)	Sediment samples	Sample method	Data	Variables	Project	References
CV10_28	GSI, MI, DCU, UCC	06/2010	✓	Day grab Box core	✓	Clay, silt, sand, water depth, total iron, TOC	INFOMAR	O'Reilly et al., 2014 Murphy et al., 2016
KRY22_03	GSI, MI, DCU, UCD, UCC, Fugro	03/2022	✓	Day grab	✓	Clay, silt, sand, water depth, TOC, total iron	PREDICT	Furey & Cullen, 2022
CE23_016	MI, DCU, UCD, NGU, AFBI	10/2023	✓	Day grab, Box core, Vibro-core, Gravity core	✓	Clay, silt, sand, water depth, total iron, TOC, LOI, Bulk density (BD)	QUEST	Grey et al., 2024; Chatting et al., 2025

Agencies/funders include Geological Survey Ireland (GSI), Marine Institute (MI), Dublin City University (DCU), University College Dublin (UCD), University College Cork (UCC), Geological Survey of Norway (NGU), Agri-Food and Biosciences Institute (AFBI), Queen's University Belfast (QUB) and the Environmental Protection Agency (EPA). Survey codes: CV10_28 (Celtic Voyager 2010–28), KRY22_03 (Keary 2022–03) and CE23_016 (Celtic Explorer 2023–16). Projects: Quantification, characterisation, source and fate of past and present carbon storage in coastal and offshore sediments for effective marine management (QUEST), Prediction of Irish coastal transformation (PREDICT), and Integrated Mapping for the Sustainable Development of Ireland's Marine Resource (INFOMAR)

2010), PAST (Hammer et al., 2001) and Excel. Summary statistics were calculated for all variables. The baseline WIS model used ordinary least squares (OLS) regression of TOC against LOI, followed by residual inspection. Structural breaks were tested using Chow tests across candidate thresholds for mud, water depth, clay, silt, sand, gravel and total Fe, with thresholds constrained to leave at least 15 samples per group. Candidate splits were ranked by F-statistic and *p*-value.

Group differences were evaluated using analysis of covariance (ANCOVA): $\text{TOC} = ([a + b] \times \text{LOI}) + (c \times \text{Group}) + (d \times [\text{LOI} \times \text{Group}])$. A significant interaction term indicated different slopes, while group effects indicated intercept differences. Final group models were assessed using regression fit, residuals and practical interpretability. Spearman correlations explored variable relationships, and Mann–Whitney *U* tests were used for non-normal group comparisons (Nachar, 2008).

Results

XRF calibration checks indicated acceptable Fe accuracy across the calibration range:

measured-versus-certified slopes were 0.97 ($R^2=0.9469$) for NCSDC 73308, 0.90 ($R^2=0.8069$) for TILL-4 and 0.99 ($R^2=0.9897$) for GBW 07411. The dataset covered wide sediment and carbon ranges (Table 2 and Table S1). LOI ranged from 0.33% to 4.95% (mean 1.23%; mean Relative standard deviation (RSD) 3.53%), and measured TOC ranged from 0.03% to 2.02% (mean 0.40%; mean RSD 2.65%). Water depths ranged from 3 to 114 m; mud content ranged from 0% to 97.66%; and Fe ranged from 0.41% to 3.78%, generally increasing in finer sediments. The paired LOI-TOC subset was smaller than the full metadata set, so external validation remains required.

Across groups, TOC correlated strongly with LOI ($r=0.85\text{--}0.98$), with the strongest relationship in mud-rich sediments ($r=0.98$; Fig. 2). Mud content correlated with TOC and LOI in the full dataset ($r=0.66$ and 0.80) and particularly in the high-mud group. Fe also correlated with TOC and LOI ($r=0.69\text{--}0.79$), especially in high-mud sediments, but these associations are interpreted as covariation with depositional setting rather than direct causation. Water depth relationships varied by group: in the full dataset, TOC and LOI correlated with water depth; in low-mud sediments, only LOI showed a weak

Table 2 Descriptive statistics for the sediment dataset. Variables include LOI, TOC, TOC/LOI, TOC/mud, water depth, grain-size fractions and Fe

Variable	<i>n</i>	Min	Max	Median	Mean	Std. error
LOI (wt.%)	77	0.33	4.95	1.02	1.23	0.09
TOC (wt.%)	98	0.03	2.02	0.21	0.40	0.04
TOC/LOI	77	0.00	0.41	0.21	0.22	0.01
TOC/Mud	86	0.00	0.30	0.02	0.03	0.00
Water Depth (m)	99	3.00	114.00	22.00	31.48	2.97
Clay (%)	97	0.00	26.36	2.82	7.07	0.88
Silt (%)	99	0.00	94.53	7.24	21.78	2.89
Mud (%)	99	0.00	97.66	9.15	25.88	3.19
Sand (%)	99	2.34	100.00	86.33	70.78	3.16
Gravel (%)	99	0.00	74.88	0.08	3.33	1.14
Fe (%)	95	0.41	3.78	0.85	1.20	0.08

Western Irish Sea											
	LOI	TOC	TOC/LOI	TOC/Mud	Depth	Clay	Silt	Mud	Sand	Gravel	Fe
LOI		0.00	0.00	0.03	0.00	0.00	0.00	0.00	0.00	0.01	0.00
TOC	0.91		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
TOC/LOI	0.51	0.83		0.29	0.00	0.00	0.00	0.00	0.00	0.00	0.00
TOC/Mud	-0.26	-0.33	-0.13		0.01	0.00	0.00	0.00	0.00	0.00	0.00
Depth	0.62	0.62	0.39	-0.29		0.00	0.00	0.00	0.00	0.00	0.00
Clay	0.72	0.84	0.63	-0.58	0.59		0.00	0.00	0.00	0.00	0.00
Silt	0.64	0.78	0.56	-0.74	0.49	0.94		0.00	0.00	0.00	0.00
Mud	0.66	0.80	0.59	-0.72	0.51	0.96	0.99		0.00	0.00	0.00
Sand	-0.71	-0.83	-0.56	0.50	-0.59	-0.87	-0.87	-0.88		0.01	0.00
Gravel	-0.29	-0.34	-0.36	0.50	-0.48	-0.50	-0.53	-0.50	0.27		0.00
Fe	0.70	0.80	0.65	0.54	0.54	0.71	0.72	0.71	-0.69	-0.49	
Lr Mud											
LOI		0.00	0.10	0.75	0.00	0.00	0.02	0.01	0.00	0.05	0.00
TOC	0.85		0.00	0.47	0.11	0.00	0.00	0.00	0.00	0.00	0.00
TOC/LOI	0.22	0.67		0.01	0.88	0.00	0.06	0.02	0.03	0.65	0.00
TOC/Mud	0.05	0.10	0.37		0.14	0.00	0.00	0.00	0.21	0.01	0.76
Depth	0.42	0.20	-0.02	0.20		0.63	0.10	0.21	0.42	0.26	0.60
Clay	0.46	0.56	0.39	-0.41	0.06		0.00	0.00	0.00	0.44	0.17
Silt	0.31	0.42	0.24	-0.79	-0.20	0.80		0.00	0.00	0.60	0.06
Mud	0.35	0.48	0.30	-0.73	-0.15	0.89	0.98		0.00	0.98	0.08
Sand	-0.51	-0.59	-0.28	0.17	-0.10	-0.69	-0.69	-0.73		0.00	0.13
Gravel	0.26	0.35	0.06	0.36	0.14	0.10	-0.06	0.00	-0.51		0.94
Fe	0.41	0.44	0.41	0.04	-0.07	0.18	0.24	0.22	-0.19	0.01	
Hr Mud											
LOI		0.00	0.07	0.25	0.22	0.02	0.02	0.02	0.02	0.04	0.00
TOC	0.98		0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.03	0.00
TOC/LOI	0.43	0.76		0.11	0.17	0.10	0.07	0.08	0.08	0.04	0.04
TOC/Mud	0.29	0.46	0.38		0.05	0.61	0.82	0.70	0.70	0.43	0.26
Depth	0.30	0.53	0.34	0.35		0.01	0.06	0.01	0.01	0.02	0.04
Clay	0.54	0.80	0.40	0.09	0.47		0.00	0.00	0.00	0.16	0.00
Silt	0.54	0.75	0.43	0.04	0.34	0.93		0.00	0.00	0.16	0.00
Mud	0.54	0.80	0.42	0.07	0.43	0.97	0.94		0.00	0.27	0.00
Sand	-0.54	-0.80	-0.42	-0.07	-0.43	-0.97	-0.94	-1.00		0.27	0.00
Gravel	-0.48	-0.40	-0.48	-0.15	-0.42	-0.26	-0.26	-0.20	0.20		0.74
Fe	0.65	0.82	0.50	0.21	0.37	0.91	0.95	0.92	-0.92	-0.06	

Fig. 2 Spearman correlation matrices for the WIS, Low mud and High mud groups. The upper triangle shows *p*-values and the lower triangle shows correlation coefficients (*r*), indicating the strength and direction of relationships among variables

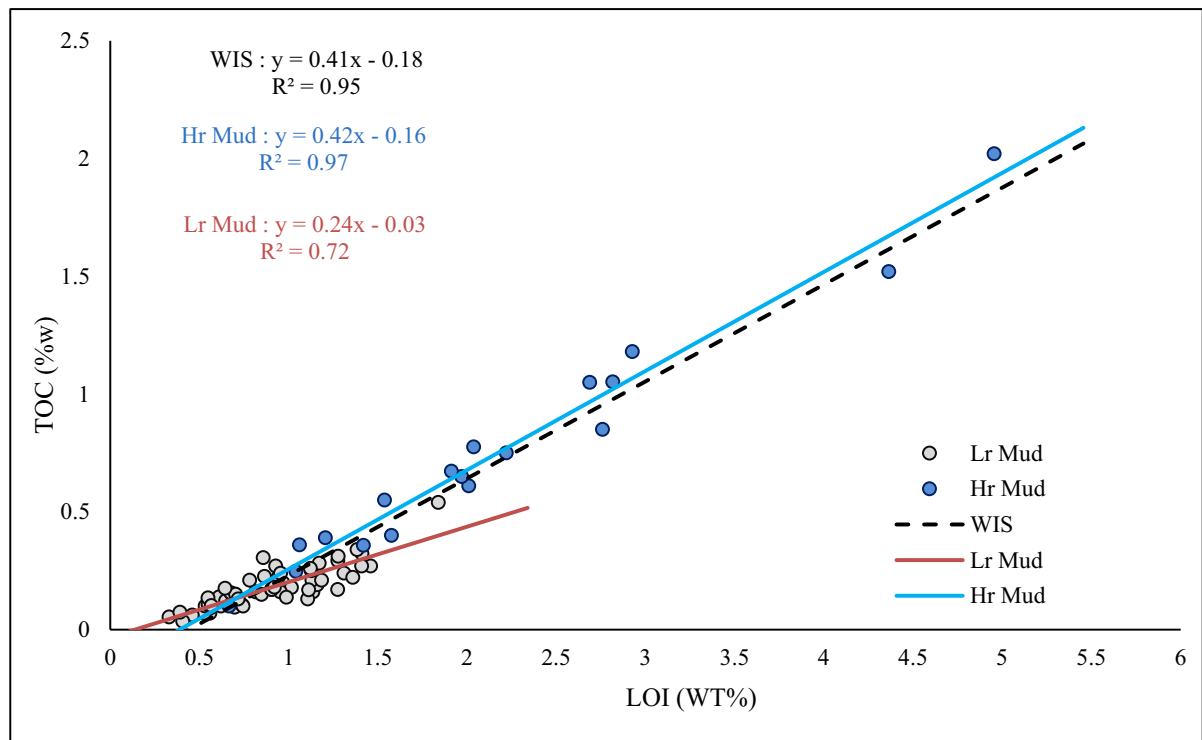


Fig. 3 Linear regression plots for the Western Irish Sea (WIS) dataset and the low mud and high mud groups. Equations and R^2 values are shown for each trend line

Table 3 Chow-test results for candidate structural breaks in the LOI-TOC relationship

Covariate	Threshold	F statistic	p	n low	n high	Slope low	Slope high
TOC/LOI	0.26	55.15	2.6×10^{-15}	57	20	0.23	0.40
Mud	19.8	32.93	6.4×10^{-11}	59	18	0.24	0.42
Fe	0.99	42.41	9.7×10^{-13}	55	18	0.22	0.41
Water depth	29.6	26.44	2.3×10^{-9}	62	15	0.26	0.41

significant water depth relationship; in high-mud sediments, TOC was more strongly water depth-associated. These patterns support testing stratified models.

LOI and measured TOC were strongly correlated across the WIS dataset (Fig. 3), yielding the baseline model:

$$\text{TOC} = 0.41 \times \text{LOI} - 0.18$$

The model explained ~95% of measured TOC variance ($R^2 = 0.9478$). The negative intercept (-0.18) indicates a small low-LOI offset, consistent with potential inorganic mass loss or methodological bias in low-TOC sands. Residuals suggested that the pooled WIS model slightly over-predicted some

sandy, low-TOC samples and under-predicted some high-TOC muds, supporting stratified analysis.

Chow tests identified the largest LOI-TOC structural breaks (Table 3). TOC/LOI gave the smallest p -value but is derived from the predictor and response variables and is therefore not a practical independent stratifier. Fe produced the strongest independent split (0.99% Fe; $p = 9.7 \times 10^{-13}$), followed by mud content (19.8% mud; $p = 6.4 \times 10^{-11}$) and water depth (29.6 m; $p = 2.3 \times 10^{-9}$). Clay, silt and sand thresholds identified substantially overlapping fine- versus coarse-sediment groups. Fe and depth are therefore interpreted as proxies for sedimentological and depositional setting rather than independently validated

Table 4 ANCOVA results for group-wise LOI-TOC models using Chow-test thresholds

Factor [Groups]	<i>n</i>	<i>F</i>	<i>p</i>	<i>F</i>	<i>p</i>	Slope		Intercept	
		[slopes]	[slopes]	Interaction	Interaction	Lower	Higher	Lower	Higher
Mud	77	10.93	1.5×10^{-3}	47.98	1.4×10^{-9}	0.24	0.42	-0.03	-0.16
Fe	77	14.98	2.3×10^{-4}	36.4	6.1×10^{-8}	0.25	0.41	-0.04	-0.14
Water depth	77	14.81	2.5×10^{-4}	31.89	2.9×10^{-7}	0.26	0.41	-0.05	-0.14

causal controls. Mud content was selected as the primary stratifier because it is statistically significant, geochemically interpretable and routinely measurable. A two-group model was therefore developed using the 19.8% mud threshold.

ANCOVA results for Fe and water depth are included because they may act as practical proxies where grain-size data are incomplete. Mann–Whitney *U* tests showed significant differences between low mud and high mud groups for all variables (Table S2).

ANCOVA indicated significant slope differences between mud, Fe and water depth groups (Table 4; Fig. 3). The high-mud slope (0.42) was nearly twice the low-mud slope (0.24), meaning that a given LOI corresponds to higher TOC in fine, deeper sediments than in shallower, sand-dominated sediments (Diesing et al., 2017). Intercepts also differed (−0.16 for high mud; −0.03 for low mud).

Near-zero LOI extrapolation should be treated cautiously because fitted intercepts fall outside the main management range and may reflect analytical or mineralogical effects. Mean TOC/LOI ratios were also higher in fine sediments (low mud=0.20; high mud=0.34), supporting sediment-class-specific conversion.

Discussion

Improving LOI-TOC conversion in marine sediments

The WIS data are not well represented by one pooled LOI-TOC relationship. Chow tests and ANCOVA revealed structural breaks associated with mud content, water depth and Fe concentration: mud-rich or Fe-rich sediments had steeper TOC/LOI slopes than coarse, Fe-poor sediments. Calibrated slopes were ~0.41 percentage points TOC per 1LOI (wt.%) in deep or mud-rich beds and ~0.24 percentage points TOC per 1 LOI (wt.%) in shallow or coarse sites. These findings are consistent with the ability of fine

mineral matrices, clay interlayers and Fe/Mn oxides to retain and protect organic matter (Barber et al., 2017; Chen et al., 2020; Johnson et al., 2015; Lalonde et al., 2012; Moore et al., 2023; Wei et al., 2025). In coarse sediments, low organic content and small absolute LOI values can make TOC estimates more sensitive to minor inorganic mass losses or seasonal/anthropogenic organic inputs (Ahrens et al., 2020; Kubo & Kanda, 2017).

The Fe split indicates a geochemical association rather than a direct causal test. Fe phases, including iron (oxyhydr)oxides and sulfides, can co-accumulate with TOC in low-energy, fine-grained or low-oxygen settings (Egger et al., 2017; Taylor & Macquaker, 2011). However, Western Irish Sea sediments are affected by bioturbation by *Nephrops norvegicus* and by trawling, with reported Irish Sea trawling intensities of 2.70–11.90 yr^{−1} and measurable benthic impacts (Jones et al., 2002; Hinz et al., 2017; O’Sullivan et al., 2014; Hiddink et al., 2016). These processes can alter sediment structure, biogeochemical cycling and OM-mineral relationships (Lønborg et al., 2024; Paradis et al., 2021). The TOC/mud ratio may help separate grain-size effects from disturbance effects, as shown for intensively trawled North Sea sediments (Zhang et al., 2024).

Fe and water depth can therefore help flag shifts in TOC/LOI where grain-size data are incomplete, but they should be used as proxies for depositional setting. Mineral associations remain important mechanistic context: nitrogen-rich OM may bind closely to Fe-rich minerals, whereas carbon-rich OM may be preserved within aggregate pores (Zhang et al., 2025), consistent with distinct mineral protection pathways for different OM pools (Blattmann et al., 2019; Yedema et al., 2024; Bauer et al., 2013; Moore et al., 2023; Barber et al., 2017; Wei et al., 2025).

Implications and recommendations

National seabed carbon inventories need reliable point data for spatial modelling (Luisetti et al., 2019,

2020). Open datasets such as MOSAIC and EURO-CARBON support harmonisation, and EURO-CARBON specifically notes that LOI-to-TOC conversion depends on local physical factors (Graversen et al., 2025; Paradis et al., 2023). A single conversion can over- or under-estimate TOC in particular sediment regimes. In this study, stratified models based on mud content, Fe or water depth improved fit within WIS groups ($R^2=0.97$ and 0.72 for high- and low-mud sediments), suggesting a practical route for converting low-cost LOI measurements when paired calibration data, sediment-class metadata and uncertainty propagation are available. Tables 5 and 6 summarise methodological and accounting recommendations; they should be treated as guidance for measurement consistency and uncertainty reporting, not as evidence that LOI alone is sufficient for national accounting.

A further implication of this study is that national sediment carbon datasets need not rely solely on standalone carbon-focused surveys. The present dataset integrates archived samples, routine geotechnical data, particle-size measurements, LOI results and targeted TOC analyses from multiple research, monitoring and seabed-mapping activities. This approach could be expanded through collaboration among national mapping programmes, research institutions, environmental regulators, local authorities, offshore renewable-energy developers, ports, fisheries-management bodies and commercial survey providers. Shared study design, sample archiving, infrastructure, metadata standards and co-funding would allow routine marine surveys and infrastructure projects to strengthen national seabed carbon assessment, improve spatial coverage, reduce duplicate sampling and support more robust carbon-stock mapping and policy evaluation.

Limitations and uncertainty

This regional case study should not be interpreted as providing a universal LOI-to-TOC conversion. The paired LOI-TOC dataset is smaller than the full metadata set, group sizes are uneven (18 high-mud and 59 low-mud samples), and the 19.8% mud, 0.99% Fe and 29.6 m water depth thresholds were selected using exploratory structural-break analysis without independent validation. Application to other regions or sedimentary settings therefore requires new paired LOI-TOC measurements to calibrate and

independently validate locally appropriate conversion relationships and any associated sediment-class thresholds. LOI provides an indirect estimate of sediment organic matter and cannot be expected to achieve the analytical specificity or accuracy of direct TOC determination by combustion-based elemental analysis. LOI at 450 °C can include organic matter, clay structural water and other volatile or mineral-associated losses. Results can also vary with ignition temperature, sample mass, sieving, drying, carbonate content and desiccation. These limitations do not preclude the use of LOI for estimating TOC, but they require conversion relationships to be empirically calibrated against direct TOC measurements and interpreted within the sedimentological and geochemical context of the samples. Fe and water depth co-vary with grain size and depositional environment in this dataset and should be treated as practical proxies unless supported by independent geochemical and mineralogical evidence. Prediction uncertainty is greatest in low-OC coarse sediments for samples close to classification thresholds.

Importantly, this study evaluates the conversion of LOI measurements to estimated TOC concentration only. It does not directly quantify sediment carbon stocks, accumulation or burial rates, carbon permanence or climate-mitigation potential. Where calibrated LOI-to-TOC relationships are applied to additional sediment samples, the resulting TOC estimates may subsequently contribute to stock calculations, but these require additional information including dry bulk density, sediment thickness or defined water depth intervals, grain-size and spatial data, together with appropriate propagation of conversion, measurement and spatial uncertainties. Estimates of accumulation or burial additionally require independent chronological constraints such as radionuclide dating.

Conclusion

This study demonstrates that the principal value of LOI-to-TOC conversion is not to replace direct elemental analysis, but to extend the spatial value of existing and future marine sediment datasets where LOI is available, but direct TOC measurements are limited. For the WIS dataset, sediment-class-specific conversion improved model fit and reduced conversion bias relative to a single pooled LOI-to-TOC

Table 5 Recommendations for methodological aspects of sediment carbon analysis

ID	Recommendation	Key implementation	Resources	Expected impact	Main risk
R1a	Measure TOC on a 10–25% calibration subset and LOI on all samples	Stratify by sediment class; pair LOI-TOC; update after protocol changes	Elemental analyser or partner lab; sampling plan	Expands coverage while retaining direct-OC traceability	Rare sediment types may be under-sampled; inter-lab offsets
R1b	Standardise and report the LOI protocol	Record drying, mass, crucible, ignition temperature/time, desiccation and replication	Muffle furnace; balance; desiccator; SOP; training	Improves comparability and auditability	Marine carbonates/clays may need documented deviations
R1c	Control carbonate removal and inorganic-carbon correction	Classify carbonate likelihood; apply one validated IC/removal SOP; record method	Elemental analyser; IC/carbonate testing capacity	Reduces systematic OC bias across carbonate gradients	Acidification can remove OC; carbonate behaviour varies
R1d	Test clay/mineral interference where fines are high	Use sensitivity tests or TGA/XRD subsets to separate mass-loss processes	Mineralogy/TGA access; specialist interpretation	Improves LOI interpretation in clay-rich sediments	Adds cost and may not yield universal corrections
R2a	Use stratified LOI-to-TOC models rather than one regional factor	Define classes; fit regressions; validate interactions; publish applicability rules	Paired LOI-TOC, grain-size/Fe/water depth metadata; statistics	Reduces regime-specific over/under-estimation	Misclassification near thresholds; model drift
R2b	Require grain-size metadata	Report mud, silt, clay, sand and method consistently; integrate into model selection	Laser diffraction or sieving/pipette; QA procedures	Supports correct stratified conversion and spatial modelling	Method offsets require harmonisation
R2c	Use Fe as a secondary proxy where grain size is absent	Measure Fe with certified references and compare Fe- versus mud-based splits	XRF/ICP/FAAS; standards; QC logging	Helps classify depositional setting when physical data are incomplete	Fe is a proxy and should not be over-interpreted causally

Table 6 Recommendations for national carbon stock accounting strategies

ID	Recommendation	Key implementation	Resources	Expected impact	Main risk
R3a	Measure dry bulk density and water content routinely	Define core/section SOP; measure volume and dry mass; propagate uncertainty	Corers, liners, oven, balance and volume tools	Improves stock accuracy and comparability	Compaction and disturbance can bias DBD
R3b	Standardise sediment depth horizons and reporting units	Use defined horizons (e.g. 0–10, 10–30, 30–100 cm); record recovery/compaction	Coring capability; database schema	Improves comparability with global benchmark	Bioturbation and deeper coring complicate interpretation
R3c	Report burial only with accumulation-rate evidence	Use $^{210}\text{Pb}/^{137}\text{Cs}$ or equivalent chronologies; calculate MAR and uncertainty	Radiometric dating; chronology expertise	Separates stock estimates from higher-tier burial evidence	Dating assumptions and mixing may dominate uncertainty
R3d	Operate a formal QA/QC system	Use replicates, blanks, CRMs, inter-lab checks, calibration logs and QC flags	CRMs; LIMS/structured data; staff time	Supports MRV credibility and detects lab drift	Coordination costs; criteria may reduce usable data
R4a	Integrate sampling and spatial modelling	Sample by strata; model with cross-validation; report uncertainty and applicability	GIS/modelling expertise; predictor layers	Improves scaling from point data to stock maps	Predictor layers and extrapolation can bias maps
R4b	Include disturbance and vulnerability layers	Link trawling/dredging/storm data to sampling strata and scenario tests	Pressure datasets; stakeholder coordination	Improves policy relevance and permanence assessment	Attribution is difficult and context dependent
R4c	Align outputs with recognised accounting frameworks	Document boundaries, tiers, activity data, uncertainty and verification traceability	Policy expertise; QA review	Makes outputs more usable for MRV and reporting	Framework mismatch may cause misinterpretation

SOP standard operating procedure, *IC* inorganic carbon, *TGA* thermogravimetric analysis, *XRD* X-ray diffraction, *ICP* inductively coupled plasma analysis, *FAAS* flame atomic absorption spectroscopy, *QA/QC* quality assurance/quality control, *CRM* certified reference material, *LIMS* laboratory information management system, *DBD* dry bulk density, *MAR* mass accumulation rate, *MRV* monitoring, reporting and verification, *GIS* geographic information system

relationship, with mud content providing the clearest basis for model separation. Direct TOC measurement nevertheless remains necessary for model development, independent validation and high-confidence inventory reporting. Future work should test whether similar sediment-class relationships occur in other regions using large datasets such as MOSAIC and EURO-CARBON, while resolving LOI method standardisation and mineralogical interference (e.g. clay structural water and carbonate loss).

Marine sedimentary blue carbon is increasingly relevant to marine protected area (MPA) assessment and marine spatial planning (Allcock et al., 2024; Bergagna et al., 2024; Laffoley & Baxter, 2022; Marine Protected Area Advisory Group, 2020; Visalli et al., 2020). Remaining gaps in dry bulk density, grain size and accumulation-rate data limit stock accuracy (Graversen et al., 2025), so national seabed mapping programmes such as INFOMAR and MAREANO (Norwegian seabed mapping programme) provide valuable sampling platforms (Bøe et al., 2022; Guinan et al., 2020). The framework presented here can increase OC data availability where LOI is the best available option. Indeed, retrospective application of validated conversion models could increase the spatial density of estimated TOC data where archived LOI measurements are accompanied by sufficient metadata, particularly grain-size and location information, but it should not be used alone to infer accumulation, burial, permanence or mitigation value. Beyond the specific LOI–TOC calibration, this study demonstrates the value of collating data from multiple marine survey, monitoring and research sources. Coordinated sample archiving, shared infrastructure, common metadata standards and targeted collaboration among mapping agencies, research institutions, regulators, local authorities and offshore infrastructure stakeholders could improve spatial coverage while reducing duplication of sampling effort. This approach would support more robust national seabed carbon assessment, monitoring and policy evaluation while merging multidisciplinary knowledge to enhance interpretation of data.

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Data availability Data will be uploaded onto a publically accessible platform.

Declarations

Ethics approval All authors have read, understood, and have complied as applicable with the statement on “Ethical responsibilities of Authors” as found in the Instructions for Authors.

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